Standard Operating Procedure for Resolving Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node YEL-ACC-17

Node: YEL-ACC-17 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6665

Description

The optical transport failure on Huawei Access Node YEL-ACC-17 at Yelahanka Sub-station is caused by the degradation of SFP/XFP transceivers or fiber cuts, resulting in a loss of optical signal power. This is due to the fact that the Huawei Access Node uses Optical Transport Network (OTN) technology to transmit data over optical fibers. The OTN protocol stack, which includes the Generalized Multi-Protocol Label Switching (GMPLS) control plane, relies on the integrity of the optical signal to ensure reliable transmission. When the SFP/XFP transceivers or fibers are degraded, the optical signal power is reduced, leading to errors and alarms in the control plane. The loss of optical signal power can be caused by various factors, including fiber cuts, SFP/XFP degradation, or excessive optical signal attenuation. In this SOP, we will diagnose and resolve the optical transport failure on Huawei Access Node YEL-ACC-17 by isolating and replacing the faulty SFP/XFP transceivers or fibers.

Pre-requisite Checks

- Verify that the redundant link is operational and available for switchover.

- Dispatch remote hands to Yelahanka Sub-station to perform physical checks on the SFP/XFP transceivers and fibers.

- Check the weather conditions and ensure that there are no external factors affecting the fiber cables.

- Gather all necessary spares and tools for the replacement of SFP/XFP transceivers or fibers.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP transceivers	High
Fiber cuts	Medium
Excessive optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

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Command: display optical-transport interface, Expected Output:
Interface Name SFP/XFP Type Optical Signal Power Receive Power
Transmit Power, Description: This command displays the optical
interface configuration and status, including the SFP/XFP type,
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optical signal power, receive power, and transmit power. It will help identify if the SFP/XFP transceivers are degraded or if there are any issues with the optical signal power.

Command: display optical-transport alarm, Expected Output: Alarm Type Alarm Description Current State Last Clear Time, Description: This command displays the optical alarm configuration and status, including the alarm type, description, current state, and last clear time. It will help identify if there are any alarms related to optical transport failures.

Command: display optical-transport fiber, Expected Output: Fiber Name Fiber Type Fiber Length Attenuation, Description: This command displays the fiber configuration and status, including the fiber name, type, length, and attenuation. It will help identify if there are any issues with the fiber cables.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceivers, Command: replace optical-transport interface, Description: Replace the degraded SFP/XFP transceivers with new ones and reconfigure the optical interface to ensure proper operation.

Step 2: Step: Repair or replace the fiber cables, Command: replace optical-transport fiber, Description: Repair or replace the faulty fiber cables and reconfigure the optical interface to ensure proper operation.

Step 3: Step: Verify the optical signal power, Command: display optical-transport interface, Description: Verify that the optical signal power is within the normal range after replacing the SFP/XFP transceivers or fibers.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output: Interface Name SFP/XFP Type Optical Signal Power Receive Power Transmit Power, Description: Verify that the optical interface configuration and status are within the normal range.

Command: display optical-transport alarm, Expected Output: Alarm Type Alarm Description Current State Last Clear Time, Description: Verify that there are no alarms related to optical transport failures.

Command: display optical-transport fiber, Expected Output: Fiber Name Fiber Type Fiber Length Attenuation, Description: Verify that the fiber configuration and status are within the normal range.
```

Roll-back Procedures

- Step: Revert the configuration, Command: undo replace optical-transport interface, Description: Revert the configuration to the previous state before replacing the SFP/XFP transceivers or fibers.

- Step: Revert the fiber configuration, Command: undo replace optical-transport fiber, Description: Revert the fiber configuration to the previous state before replacing the fiber cables.